

Weekly Problem 6

Ravi Kini

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Exercise 1

Let there be a Kripke model with moments 0, 1, and 2 where $0 \leq 1$ and $0 \leq 2$. Let p be verified at $\{0, 2\}$ and q be verified at $\{2\}$.

For $p \Rightarrow q$ to be verified at 0, for all v such that $0 \leq v$, if p is verified at 0, then q is verified at v . However, we see that $0 \leq 1$, but p is verified at 0 and q is not verified at 1. Consequently, $p \Rightarrow q$ is not verified at 0.

For $\neg\neg p$ to be verified at 0, for all v such that $0 \leq v$ and all w such that $v \leq w$, p is verified at w . Equivalently, by transitivity, for all v such that $0 \leq v$, p is verified at v . However, we see that $0 \leq 1$, and p is not verified at 1. Consequently, $\neg\neg p$ is not verified at 0.

For $\neg p \vee q$ to be verified at 0, either p is not verified at 0 or q is verified at 0. However, we see that p is verified at 0 and q is not verified at 0. Consequently, $\neg p \vee q$ is not verified at 0.